

		for the field lines. At the middle, the field lines due to X and Y are both positive (i.e. towards the right), so they add up.
20	D	As voltage across the lamp increases, its temperature increases, leading to more vigorously vibration of lattice ions. The rate of collision between free electrons and lattice ions increases leading to increases in resistance.
21	B	Since the length of the wire is connected into the circuit, the current passing through the two cross-sections R and S is the same i.e. the rate of flow of charge is the same